

Day 7 Homework

Choose a retrieval pipeline for my own documents

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The short version. I built a chatbot over 23 made-up IT help desk documents. When someone asks a question, the chatbot first has to *find the right piece of text* (this is called **retrieval**) and then write an answer from it. If retrieval picks the wrong text, the answer will be wrong. So this homework tests only the finding step.

I wrote 20 test questions and tried four ways of finding text. The most careful way, called **full**, found the right text for all 20 questions. It is slower and costs two extra model calls per question, so I chose it because users of a chat often ask follow-up questions, and only full handles those well.

The four ways of searching, in plain English

Mode	What it does
vector	Searches by meaning. It turns the question and every piece of text into numbers and picks the closest ones. Fast, but can blur exact words such as error codes.
hybrid	Vector search plus ordinary keyword search, with the two result lists combined. Good for exact words and codes. No extra model cost.
rerank	Hybrid first, then an AI model reads the top 10 results and re-orders them by how well they answer the question. One extra model call.
full	The AI model first rewrites the question into a clear search (for example, it fills in what 'that' refers to), then does hybrid and rerank. Two extra model calls.

How to read the scores

Word	Meaning
hit@4	Out of 20 questions, how many times the right text was among the 4 pieces handed to the chatbot. Higher is better. 90% means 18 of 20.
MRR	How close to the top the right text was. 1.00 would mean always first. Higher is better.
ms / calls	Time per question in milliseconds, and how many AI model calls retrieval needed. Lower is cheaper and faster.

1. Test set: 20 questions

Each question has the file that answers it and a short **evidence phrase** copied word for word from that file. A question counts as found only if one of the 4 retrieved pieces comes from the right file *and* contains the evidence phrase. The mix is: 6 plain, 4 reworded (my words, not the document's), 4 exact code (error codes), 3 follow-up (vague without the previous question) and 3 messy (chatty, as people really write). The file is data/my_retrieval_test_set.csv.

#	Kind	Question	Previous question	Expected file	Evidence phrase
1	plain	How do I set up MFA on a new phone?		mfa_setup_guide.md	install Northwind Authenticator from the App Store
2	plain	How long do you keep daily server backups?		backup_and_recovery_runbook.md	kept for 30 days
3	plain	My VPN keeps cutting out at home. What can I try?		vpn_troubleshooting.md	Switch the protocol in NorthLink settings
4	plain	What laptop does a new starter get?		new_starter_checklist.md	14-inch standard laptop, a headset
5	plain	I forgot my password and cannot sign in. What do I do?		password_policy.md	self-service page at reset.northwind.example
6	plain	How big a file can I email to a colleague?		email_and_calendar_faq.md	25 MB. For larger files
7	reworded	How early should I refresh the security certificate on my work laptop for remote access?		Day06_Slide15_policy_vpn_remote_access.pdf	at least 14 days before it expires
8	reworded	If I accidentally delete a document, how long can I get it back?		file_sharing_and_storage.md	stay in the recycle bin for 93 days
9	reworded	Am I allowed to paste customer information into a public chatbot?		Day06_Slide15_policy_data_handling.pdf	must never be pasted into public AI tools
10	reworded	What should I do if my laptop's battery looks puffed up?		Day06_Slide15_policy_equipment_and_use.pdf	stop using the laptop immediately
11	exact code	What does error VPN-809 mean?		Day07_Slide05_kb_error_codes.pdf	VPN certificate on this laptop has expired
12	exact code	How do I fix NET-2240?		Day07_Slide05_kb_error_codes.pdf	run ipconfig /flushdns from a command prompt
13	exact code	I keep getting WIFI-117, what now?		Day07_Slide05_kb_error_codes.pdf	Guest Wi-Fi access lasts 8 hours
14	exact code	What is VPN-820?		Day07_Slide05_kb_error_codes.pdf	endpoint agent on this device is missing or not reporting as healthy
15	follow-up	How do I fix that if I'm not in the office?	What does error VPN-809 mean?	Day07_Slide05_kb_error_codes.pdf	call the service desk to renew it remotely
16	follow-up	And for P3?	What is the first response time for a P1 ticket?	Day06_Slide15_policy_it_support_slas.pdf	P3: response within 4 hours

#	Kind	Question	Previous question	Expected file	Evidence phrase
17	follow-up	What if I lose the phone instead?	How do I set up MFA on a new phone?	mfa_setup_guide.md	Call the service desk on extension 4357 straight away
18	messy	hey, quick one, i've had my laptop forever and it's slow lol. is there like a schedule for swapping them out, and does IT warn me first??		laptop_standards.md	about 2 months before your replacement
19	messy	hello, i'm stuck at an airport hotel and the vpn just wont connect, says cannot reach server or something, any ideas thanks		vpn_troubleshooting.md	firewall or hotel/airport network is blocking the VPN
20	messy	Morning! got a weird email asking me to confirm my password, i did click the link already, panicking a bit, what do i do?		phishing_and_security_reporting.md	If you already clicked a link or entered your password

Output of the checker (scripts/check_test_set.py)

The checker confirms every evidence phrase really appears in its file. A wrong phrase would make retrieval look worse than it is. All 20 phrases were found.

Checked 20 questions against data/my_docs

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OK    How do I set up MFA on a new phone?
OK    How long do you keep daily server backups?
OK    My VPN keeps cutting out at home. What can I try?
OK    What laptop does a new starter get?
OK    I forgot my password and cannot sign in. What do I do?
OK    How big a file can I email to a colleague?
OK    How early should I refresh the security certificate on my wo
OK    If I accidentally delete a document, how long can I get it b
OK    Am I allowed to paste customer information into a public cha
OK    What should I do if my laptop's battery looks puffed up?
OK    What does error VPN-809 mean?
OK    How do I fix NET-2240?
OK    I keep getting WIFI-117, what now?
OK    What is VPN-820?
OK    How do I fix that if I'm not in the office?
OK    And for P3?
OK    What if I lose the phone instead?
OK    hey, quick one, i've had my laptop forever and it's slow lol
OK    hello, i'm stuck at an airport hotel and the vpn just wont c
OK    Morning! got a weird email asking me to confirm my password,
```

20 of 20 evidence phrases found.

2. Comparison of the four pipelines

I ran all 20 questions through each mode with `scripts/compare_modes.py`. This is the saved table (`data/retrieval_comparison.csv`):

Mode	hit@4	MRR	plain	reworded	exact code	follow-up	messy	avg ms	model calls
vector	90%	0.81	6/6	4/4	4/4	1/3	3/3	2154	0
hybrid	90%	0.70	5/6	4/4	4/4	2/3	3/3	2939	0
rerank	95%	0.83	6/6	4/4	4/4	2/3	3/3	6817	1
full	100%	0.77	6/6	4/4	4/4	3/3	3/3	11699	2

What the table shows

- **Plain, reworded, exact code and messy questions** were found almost equally well by every mode. Hybrid lost one plain question (*What laptop does a new starter get?*), and the other three modes found it.
- **Follow-up questions are where the modes differ.** Vector found 1 of 3, hybrid and rerank 2 of 3, and full found all 3. Full is the only mode that rewrites the question using the one before it.
- **Rerank has the best MRR (0.83):** it moves the right text nearer the top, for one extra model call.
- **Full is the most complete (100%)** but the slowest (about 11.7 seconds per question here) and its MRR (0.77) is a little lower than rerank, because the right text is found but not always first.
- Each step fixed the kind of question it was built for, apart from hybrid, which did not help on my documents: my exact-code questions already worked with plain vector search because the error code is written in the question and in the text.

Questions still missed in full mode (`scripts/eval_retrieval.py --mode full`)

None. All 20 questions were found. The full per-question results are in `data/retrieval_results_full.csv`.

3. Diagnosis: why two questions failed

Full mode missed nothing, so I followed the homework instruction for that case: I took the two questions that **vector mode missed** and used `scripts/explain_question.py` to see what each mode retrieved. Both are follow-up questions. (The complete tool output is saved in `logs/explain_output.txt`.)

Question A: "And for P3?" (previous question: "What is the first response time for a P1 ticket?")

What I saw: Vector found the support policy's first page and the service desk table, but not page 2 of the policy, which holds the sentence "P3: response within 4 hours". Hybrid, rerank and full did find page 2 (at rank 4, 3 and 3).

Cause: the question needed context. "And for P3?" says nothing about response times on its own.

Fix: rewriting. Full mode turned it into "What is the first response time for a P3 ticket?" before searching.

Question B: "How do I fix that if I'm not in the office?" (previous question: "What does error VPN-809 mean?")

What I saw: Vector, hybrid and rerank returned unrelated text about printers, Wi-Fi and email. The words "that" and "the office" give no topic. Only full mode, which rewrote it as "How to fix error VPN-809 remotely not in office", found the error-code text that says "call the service desk to renew it remotely".

Cause: the question needed context (the follow-up refers to an error code mentioned earlier).

Fix: rewriting.

Two honest notes. (1) The rank of that VPN chunk was 3 in the full evaluation run but 1 when I re-ran the explain tool, because the AI model's scores change slightly from run to run. It was found both times. (2) In that run the reranker gave the correct chunk only 1 out of 10, because the chunk starts in the middle of another error code's entry. So reranking can make a mistake when a piece of text starts mid-topic.

4. Decision record

This is my RAG_DECISION_day07.md, shown here in full. Every number comes from the table in section 2.

Decision: retrieval pipeline for the Northwind IT help desk chatbot

Context. 23 made-up Northwind IT documents (62 chunks): policies as PDFs, guides and FAQs as Markdown, and an error-code reference. The people asking are Northwind staff with everyday IT problems, often as chat follow-ups such as "And for P3?". **My target:** the right text found for at least 95% of questions (hit@4 of 95% or more), every kind of question working, and retrieval within 15 seconds on this laptop. The 15 seconds is generous on purpose: just turning the question into numbers takes about 2 seconds on this laptop, and a chatbot is not a search box.

Options. vector 90% (MRR 0.81), hybrid 90% (0.70), rerank 95% (0.83), full 100% (0.77). The full table is in section 2.

Decision. RAG_MODE=full. It is the only mode that finds the right text for all 20 questions, including all 3 follow-ups, and it meets the time limit. Rewriting is what fixed the follow-up questions, and follow-ups are normal in a chat.

Consequences (what gets worse).

- 2 extra model calls per question (one to rewrite, one to rerank), on top of the call that writes the answer.
- About 11.7 seconds per question to retrieve, compared with about 2.2 seconds for vector. Timings vary from run to run.
- MRR is lower than rerank (0.77 against 0.83): the right text is found, but sometimes lower in the top 4.
- Only 20 questions, so one question is worth 5 points. Rerank (95%) and full (100%) differ by a single follow-up question.
- Some documents contradict each other (the Markdown and PDF versions give different password expiry, lockout time and service desk hours). Search cannot fix that. The documents themselves need fixing.

Review when. The documents change, the AI model or the embedding model changes, retrieval takes more than 15 seconds, or the test set grows and the gap between rerank and full changes.

The rule I added to CLAUDE.md

"**Retrieval changes are measured:** any change to retrieval (chunking, search mode, reranking, rewriting or the number of chunks) must be measured with scripts/compare_modes.py before and after, and the numbers recorded."

In plain words: before I change how the chatbot searches, I run the comparison, change it, run it again, and write down both sets of numbers, so I know whether the change really helped.

Where everything is

All files are in the GitHub repository mithransadasivam/FDE, folder **HW #7**: the test set, the scripts' results (data/), the decision record, the CLAUDE.md rule, and the code with its 120 passing tests.